

Name _____

Accelerated Geometry Semester 1 Unit 2

Review for Test #2

Match the description below with its equation.

a. $y = -\frac{1}{4}x + \frac{3}{2}$

d. $y = 5x - 14$

b. $y = 12x - 91$

e. $y = 12x + 5$

c. $y = \frac{1}{4}x + 24$

f. $y = 5x + 5$

B 1. line passing through the point (8, 5) that is perpendicular to the line $y = -\frac{1}{12}x + 8$

A 2. perpendicular bisector of $\overline{PQ}$ with endpoints $P(4, -8)$ and $Q(8, 8)$

D 3. line passing through the point (4, 6) that is parallel to the line $y = 5x + 3$

~~Determine which of the lines are parallel and which of the lines are perpendicular.~~

$$\begin{aligned} 6 &= 5(4) + b \\ 6 &= 20 + b \\ -14 &= b \end{aligned}$$

$$y = 5x - 14$$

$$0 = -\frac{1}{4}(6) + b$$

$$0 = -\frac{3}{2} + b$$

Slope of $\overline{PQ}$

$$\frac{8 - -8}{8 - 4} = \frac{16}{4} = 4$$

Write an equation of the line passing through point P that is parallel to the given line.

5. $P(2, -3), y = -6x + 10$ $m = -6$

$$-3 = -6(2) + b$$

$$-3 = -12 + b$$

$$9 = b$$

Parallel $\rightarrow m = -6$

ans: $y = -6x + 9$

Find the distance from point Q to the given line.

6. $Q(-10, -4), 5x - y = 6$

$$-y = -5x + 6$$

$$y = 5x - 6$$

original

perpend. $m = -\frac{1}{5}$

$$-4 = -\frac{1}{5}(-10) + b$$

$$-4 = 2 + b$$

$$-6 = b$$

perpend. line
 $y = -\frac{1}{5}x - 6$

② $5x - 6 = -\frac{1}{5}x - 6$

$$5.2x = 0$$

$$x = 0$$

$$(0, -6)$$

③

$$\begin{aligned} &\sqrt{(-10 - 0)^2 + (-4 - -6)^2} \\ &\sqrt{(-10)^2 + (2)^2} \\ &\sqrt{104} \end{aligned}$$

$$10.198 \rightarrow \approx 10.2 \text{ un}$$

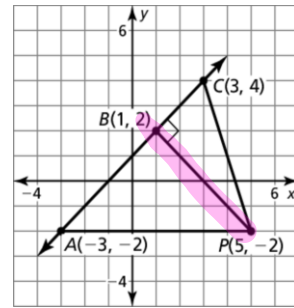
7. Find the distance from point P to $\overline{AB}$.

$$\sqrt{(1-5)^2 + (2-(-2))^2}$$

$$\sqrt{(-4)^2 + (4)^2}$$

$$\sqrt{32}$$

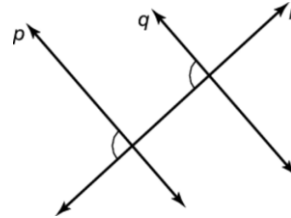
$$\approx 5.66 \text{ un}$$



8. Decide whether there is enough information to prove that $p \parallel q$.

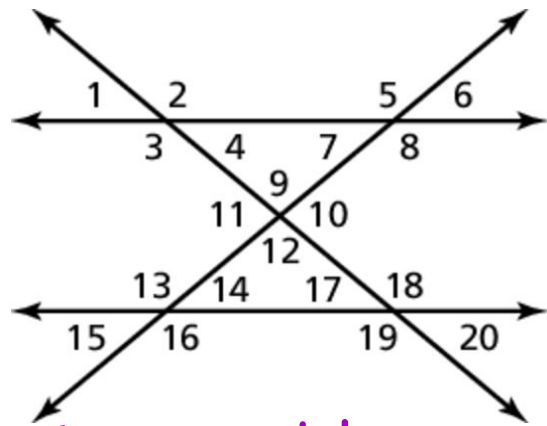
If so, state the theorem you would use.

Yes, Corresponding Angle Converse Thm



9. List one answer for each.

- $\angle 1$ and 17 are corresponding angles.
- $\angle 13$ and 5 are corresponding angles.
- $\angle 14$ and 8 are consecutive interior angles.
- $\angle 4$ and 18 are consecutive interior angles.
- $\angle 7$ and 14 are alternate interior angles.
- $\angle 17$ and 4 are alternate interior angles.
- $\angle 6$ and 15 are exterior exterior angles.
- $\angle 18$ and 15 are alternate exterior angles.

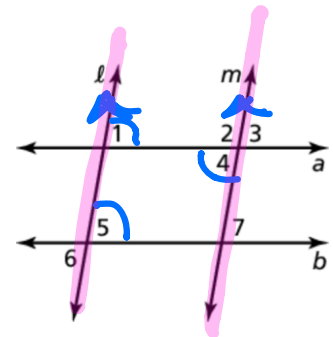


Some problems have multiple answers

10. Write a two-column proof.

Given $\ell \parallel m$ and $\angle 5 \cong \angle 4$

Prove $a \parallel b$



Statements	Reasons
① $\ell \parallel m$ $\angle 5 \cong \angle 4$	① given
② $\angle 4 \cong \angle 1$	② Alternate Int $\angle$ Thm
③ $\angle 5 \cong \angle 1$	③ transitive Prop. $\cong$
④ $a \parallel b$	④ Corresponding Angle Converse Thm

OR

② $\angle 5 \cong \angle 7$

③ $\angle 4 \cong \angle 7$

④ $a \parallel b$

② Corresponding Angle Thm

③ Transitive Prop. $\cong$

④ Alternate Int. $\angle$ Converse
Thm